Plane Curves: Singularities, Intersection Multiplicities, and Normalization

N-20240608

A polynomial equation in two variables draws a curve, but the equation retains information that may disappear in the picture. A tangent line may touch at two algebraic orders, a crossing point may contain two distinct local branches, and two curves whose affine drawings meet only once can have another intersection at infinity. The coordinate ring, local intersection multiplicity and normalization recover precisely this lost information.[1]

§1.1 Curved Images of an Affine Line

Work over the complex numbers. Consider the parabola

$$C = V(y - x^2) \subset \mathbb{A}^2_{\mathbb{C}}.$$

Its coordinate ring is

$$\mathbb{C}[C] = \mathbb{C}[x, y]/(y - x^2) \cong \mathbb{C}[x].$$

The isomorphism sends the class of y to x^2: every polynomial function on the parabola is uniquely represented by a polynomial in the x-coordinate alone.

Geometrically this comes from the parametrization

$$t \longmapsto (t, t^2)$$

with inverse given by projection onto the first coordinate. Although the parabola bends in the surrounding plane, it is isomorphic, as an affine algebraic variety, to the affine line.

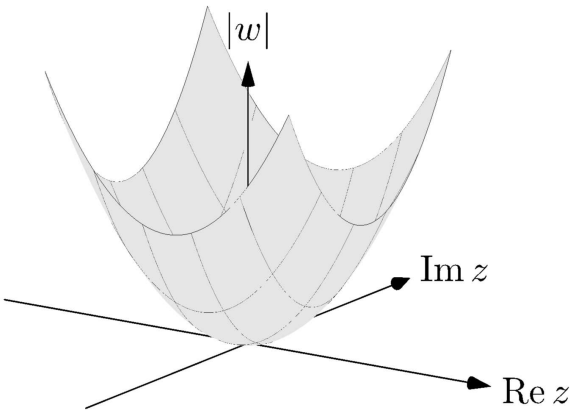


Figure 1.1: The parabola is an affine algebraic copy of the line; its apparent curvature depends on its embedding into the plane.

This distinction matters whenever a geometric claim is meant to survive a change of coordinates: being a parabola in a chosen plane is different from having the intrinsic function theory of $\mathbb{A}^1$.

[1] M. Artin, MIT 18.721, *Introduction to Algebraic Geometry*, Chapters 1 and 4; Stacks Project, §43.16.

§1.2 Tangent Cones and Singularities

For a plane curve $F(x,y) = 0$, a point $p = (a,b)$ is smooth when at least one of $F_x(p), F_y(p)$ is nonzero. The linear part

$$F_x(p)(x-a) + F_y(p)(y-b)$$

then defines its unique tangent line.

At a singular point both first derivatives vanish. Translate the point to the origin and write

$$F(x,y) = F_m(x,y) + F_{m+1}(x,y) + \cdots,$$

where each F_j is homogeneous of degree j and $F_m \neq 0$. The equation

$$F_m(x,y) = 0$$

is the tangent cone; the integer m is the multiplicity of the plane curve at the point.

For the cuspidal cubic

$$C_{\text{cusp}}: \quad y^2 = x^3,$$

the lowest term is y^2. Its tangent cone is the line $y = 0$ counted twice. For the nodal cubic

$$C_{\text{node}}: \quad y^2 = x^2(x+1),$$

the lowest terms are $y^2 - x^2 = (y-x)(y+x)$. Its tangent cone has two distinct lines, $y = x$ and $y = -x$.

Both points have multiplicity two, yet their local branches behave differently. The tangent cone makes the distinction before we attempt to draw either curve.

§1.3 Normalization separates a cusp from a node

The cusp has a polynomial parametrization

$$\nu_{\text{cusp}} : \mathbb{A}^1 \longrightarrow C_{\text{cusp}}, \qquad t \longmapsto (t^2, t^3).$$

On coordinate rings, this identifies

$$A = \mathbb{C}[x,y]/(y^2 - x^3) \cong \mathbb{C}[t^2, t^3] \subset \mathbb{C}[t] = B.$$

The element t belongs to the fraction field of A, since $t = y/x$ away from $x = 0$. It is integral over A, because it satisfies the monic equation

$$T^2 - x = 0.$$

Thus B is a finite integral extension of A. Since $\mathbb{C}[t]$ is integrally closed and has the same fraction field, it is the normalization of A. The map is one-to-one on complex points; the cusp point corresponds to $t = 0$, but the local ring becomes regular after normalization.

The node has a different parametrization:

$$\nu_{\text{node}}(t) = (t^2 - 1,\ t(t^2-1)).$$

Indeed,

$$\big(t(t^2-1)\big)^2 = (t^2-1)^2\big((t^2-1)+1\big).$$

Again $t = y/x$ away from $x = 0$, and $t^2 = x + 1$ shows that the normalization is $\mathbb{C}[t]$. This time, however,

$$\nu_{\text{node}}(1) = \nu_{\text{node}}(-1) = (0,0).$$

Two distinct points of the normalized line are identified at the node. Normalization therefore resolves the singularities in different ways: it restores a regular local parameter at the cusp and separates two branches at the node.

§1.4 Intersection Multiplicity

Two algebraic curves can meet at a single point and still contribute more than one to their intersection number. For plane curves without a common component, a local algebraic measure is

$$I_p(F,G) = \dim_{\mathbb{C}} \frac{\mathbb{C}[[x-a, y-b]]}{(F,G)}$$

at an isolated intersection $p = (a,b)$, when this quotient is finite-dimensional.

Take the cusp $F = y^2 - x^3$. Intersect it with the horizontal tangent $G = y$ at the origin:

$$\frac{\mathbb{C}[[x,y]]}{(y^2 - x^3, y)} \cong \frac{\mathbb{C}[[x]]}{(x^3)}.$$

The quotient has basis $1, x, x^2$, so

$$\boxed{I_{(0,0)}(C_{\text{cusp}}, V(y)) = 3.}$$

By contrast, intersecting with the vertical line $x = 0$ gives

$$\frac{\mathbb{C}[[x,y]]}{(y^2 - x^3, x)} \cong \frac{\mathbb{C}[[y]]}{(y^2)},$$

whose dimension is two:

$$\boxed{I_{(0,0)}(C_{\text{cusp}}, V(x)) = 2.}$$

The two lines meet the same visible point, but the tangent line has order-three contact. Under the parametrization $(x,y) = (t^2, t^3)$, this is especially transparent: the functions x and y vanish to orders two and three in t.

§1.5 Projective closure remembers a missing intersection

Homogenizing $y = x^2$ produces the projective conic

$$\overline{C}: \quad YZ = X^2 \quad \subset \mathbb{P}^2_{\mathbb{C}}.$$

The affine chart $Z = 1$ recovers $y = x^2$. Consider the vertical line $X = 0$. In affine coordinates, it meets the parabola at the origin $(0,0)$. In projective coordinates, the simultaneous equations

$$X = 0, \qquad YZ = 0$$

give two points:

$$[0:0:1] \qquad \text{and} \qquad [0:1:0].$$

The second lies on the line at infinity $Z = 0$.

Both intersections are transverse. At the affine point, use the chart $Z = 1$ with equation $y = x^2$; the line $x = 0$ cuts it simply. At the point at infinity, use the chart $Y = 1$, where $z = x^2$, and again the line $x = 0$ cuts simply. Their multiplicities are both one.

Bézout's theorem predicts the total intersection number

$$\deg(\overline{C}) \deg(V(X)) = 2 \cdot 1 = 2.$$

The projective calculation finds exactly those two intersections. The apparently missing contribution in the affine plane was at infinity, not an arithmetic error.

A different line gives another distribution of the same total count. The line $Y = 0$ is tangent to the conic at $[0:0:1]$. In the chart $Z = 1$, the quotient by $y = 0$ becomes $\mathbb{C}[[x]]/(x^2)$, so that one point contributes multiplicity two. The total remains two, but two simple intersections have merged into one tangent intersection.

§1.6 Elimination sees the same multiplicity in one variable

If $F(x,y)=0$ and $G(x,y)=0$, the resultant

$$\mathrm{Res}_y(F,G)$$

eliminates y and produces a polynomial condition on x. It can reveal possible intersection coordinates, with multiplicities and extraneous factors that must be interpreted according to the leading coefficients and the chosen projection.

For example, take the parabola $F=y-x^2$ and the horizontal line $G=y$. Eliminating y gives, up to a nonzero scalar,

$$\mathrm{Res}_y(y-x^2,y)=x^2.$$

The double root at $x=0$ is exactly the local tangency multiplicity found from the quotient ring. For the cusp y^2-x^3 intersecting $y=0$, elimination similarly gives x^3, reflecting the triple contact.

The resultant is a computational shadow of intersection theory: it remembers how many roots are merged by projection, while local rings explain the precise geometric meaning of those multiplicities.

A plane curve is therefore best studied through several cooperating descriptions. Its coordinate ring specifies polynomial functions intrinsic to the curve. The tangent cone detects the first nonzero local geometry at a singularity. Normalization recovers smooth local parameters and may separate branches. Intersection multiplicities measure tangencies algebraically, and projective completion accounts for contributions invisible in an affine drawing. Each additional tool answers a concrete question left unresolved by the preceding one.